

St Catherine's School

Year: 12
Subject: Extension I Mathematics
Time Allowed: 2 hours
(plus 5 mins reading time)
Date: August 2001

Exam number: _____

Directions to candidates:

- All questions are to be attempted.
- All questions are of equal value.
- All necessary working must be shown in every question.
- Full marks may not be awarded for careless or badly arranged work.
- Each question attempted should be started on a new booklet.
- Approved calculators and geometrical instruments are required.
- This page is a cover sheet for Section A. Write a cover page for Section B and C and include your number.
- Hand in your work in 3 bundles:
Section A Questions 1, 2 and 3.
Section B Questions 4 and 5
Section C Questions 6 and 7.

TEACHER'S USE ONLY	
Total Marks	
A	
B	
C	
TOTAL	

YEAR 12 EXTENSION 1 AUGUST 2001

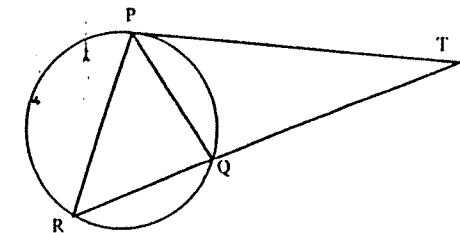
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Question 1:

- (a) Solve for x : $\frac{3}{x+5} \leq 1$ (2 marks)
- (b) A root of $e^x - x^2 = 0$ lies near $x = -0.5$. Use Newton's Method once to find a better approximation. (3 marks)
- (c) Consider the function $f(x) = 2 \sin^{-1} \frac{x}{2}$.
- (i) Find the exact value of $f(\sqrt{2})$. (1 mark)
- (ii) What is the domain and range of $f(x)$. (2 marks)
- (iii) Sketch $f(x)$. (1 mark)
- (iv) Find the equation of the tangent to the curve at $x = \sqrt{2}$. (3 marks)

Question 2:

- (a) Solve for x : $2|x-1| = 4x-1$. (3 marks)
- (b) If α, β, γ are the roots of $x^3 - 3x^2 - 2x + 1 = 0$, find:
- (i) $\alpha + \beta + \gamma$ (1 mark) (ii) $\alpha\beta\gamma$ (1 mark)
- (iii) $\alpha\beta + \alpha\gamma + \beta\gamma$ (1 mark) (iv) $\alpha^2 + \beta^2 + \gamma^2$ (2 marks)
- (c) PT is a tangent to the circle PRQ. RQ is a secant intersecting the circle in Q and R. The line RQ intersects PT at T.



(3 marks)

Question 3:

(a) Find $\int x^2(x^3 - 5)^5 dx$ using the substitution $u = x^3 - 5$ (3 marks)

(b) The angle between the lines $y = mx$ and $y = \frac{1}{7}x$ is 45° . Find two possible values of m . (3 marks)

(c) The rate at which a body cools in air is proportional to the difference between its temperature T and the temperature C of its surroundings. That is:

$$\frac{dT}{dt} = -k(T - C) \text{ where } t \text{ is the time in hours and } k \text{ is a positive constant.}$$

i) Show that $T = C + Ae^{-kt}$ is a solution to the differential equation above (where A is a real number). (2 marks)

A heated piece of metal is initially 90°C but cools to 70°C in one hour. Given the surroundings are 25°C find:

ii) the constants A and C (3 marks)
iii) the temperature of the metal after three hours (to the nearest degree). (1 mark)

Question 4:

(a) Find $\int \cos^2 2x dx$ (2 marks)

(b) A team of 4 is to be chosen from 5 boys and 6 girls. How many teams are possible if:

(i) there are no restrictions (1 mark)

(ii) the shortest girl must be included (1 mark)

(c) Show by Mathematical Induction that the following statement: (4 marks)

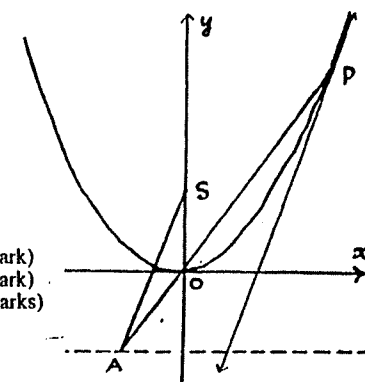
$$(1 \times 2) + (2 \times 3) + (3 \times 4) + \dots + (n-1)(n) = \frac{(n-1)n(n+1)}{3}$$

is true for all integers $n \geq 2$.

(d) The diagram shows the parabola with parametric coordinates $x = 2ap$ and $y = ap^2$. P is a point on the parabola. S is the focus and A is a point on the directrix.

The straight line drawn from point $P(2ap, ap^2)$ on the parabola through the vertex at $O(0,0)$ intersects the directrix at A .

- (i) Find the equation of line PO (1 mark)
(ii) Show that the coordinates of A are $(-\frac{2a}{p}, -a)$ (1 mark)
(iii) Prove that AS is parallel to the tangent at P . (2 marks)



Question 5:

(a) Eight different coloured beads are arranged so that two particular colours are next to each other. In how many ways can they be arranged in: (You may leave your answer in factorial notation)

- (i) a line (1 mark)
(ii) a circle (1 mark)
(iii) a necklace (1 mark)

(b) Find the exact value of

$$\int_0^2 \frac{dx}{\sqrt{4-x^2}}$$

(2 marks)

(c) Kelly throws a stone at an angle of elevation of α from the top of a tower 80 m high at an initial velocity of 50 m/sec, as in the diagram.

The acceleration due to gravity is assumed to be 10 m/sec^2 . Take the origin to be the base of the tower.

(i) Given that $\frac{d^2x}{dt^2} = 0$ and $\frac{d^2y}{dt^2} = -10$

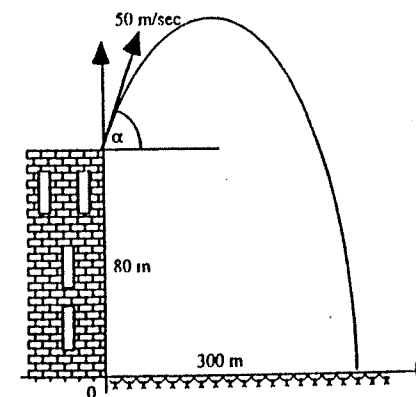
show that

$$x = 50t \cos \alpha \text{ and}$$

$$y = -5t^2 + 50t \sin \alpha + 80,$$

where x and y are the horizontal and vertical displacements of the stone in metres from the origin at time t seconds after throwing.

(ii) Kelly wants the stone to land in the sea 300 metres from the base of the tower.



(3 marks)

Question 6:

- (a) Differentiate $f(x) = \ln(\tan^3 x)$ (2 marks)
- (b) Consider the function $f(x) = \frac{x^2}{x^2 - 2}$
- (i) Give the equations of any horizontal and vertical asymptotes. (2 marks)
- (ii) Find the x and y intercepts if they exist. (1 mark)
- (iii) Given that this curve has only one stationary point and it is a local maximum, find its coordinates. (2 marks)
- (iv) Sketch the curve, indicating on your sketch all important features. (2 marks)
- (c) Show that the equation $2x^3 - 5x^2 + 3x - 2 = 0$ has only one real root. (3 marks)

Question 7:

- (a) The displacement x metres of a particle from the origin is in simple harmonic motion and is given by $x = 5 \cos \pi t$, where the time t is in seconds.
- (i) What is the period of the oscillation? (1 mark)
- (ii) What is the speed v of the particle as it moves through the origin? (2 marks)
- (b) Show that $(x-1)(x-2)$ is a factor of $P(x) = x^m(2^n - 1) + x^n(1 - 2^m) + 2^n - 2^m$ where m, n are positive integers. (2 marks)

Question 7 (continued):

- (c) An egg timer has the same shape as the curve $y = x^3$ rotated about the y axis. The top half of the egg timer is filled with sand to a depth of h units.

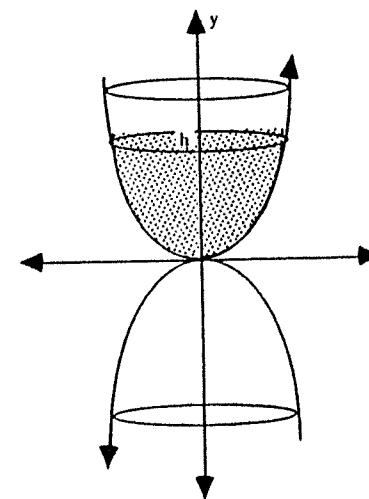
- (i) Show that the volume V of sand needed is given by $V = \frac{3\pi}{5} \sqrt{h^5}$. (3 marks)

The rate with which the sand falls into the bottom of the timer is found to be proportional to the height h of the sand in the top of the egg timer.

(ie $\frac{dV}{dt} = kh$ where k is a real number)

- (ii) Find the exact rate at which the height of the sand in the top of the egg timer is falling when $h = \frac{27}{8}$ cm if the sand is flowing through the neck at $1 \text{ cm}^3 / \text{minute}$ when $h = 5$ cm

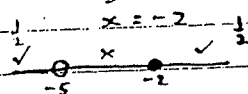
(4 marks)



1. a) $\frac{3}{x+5} \leq 1$

$x \neq -5$
Solve eqn to find critical pts

$\frac{3}{x+5} = 1$
 $3 = x+5$



$\therefore x < -5, x \geq -2$ ✓✓

b) let $f(x) = e^x - x^2$

$f'(x) = e^x - 2x$

$x_2 = x_1 - \frac{f(x_1)}{f'(x_1)}$

$= -0.5 - \frac{f(-0.5)}{f'(-0.5)}$

$= -0.5 - \frac{e^{-0.5} - (-0.5)^2}{e^{-0.5} - 2(-0.5)}$

$= -0.5 - \frac{0.35653...}{1.60653...}$

$= -0.72196...$ ✓✓

c) i) $f(\sqrt{2}) = 2 \sin^{-1} \frac{\sqrt{2}}{2} = \frac{\pi}{2}$

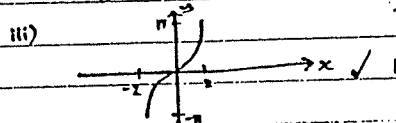
$= 2 \sin^{-1} \frac{1}{\sqrt{2}}$

$= 2(\frac{\pi}{4})$

$= \frac{\pi}{2}$ ✓✓

ii) Domain: $-2 \leq x \leq 2$ ✓✓

Range: $-\pi \leq y \leq \pi$ ✓✓



iv) If $x = \sqrt{2}$, $f(\sqrt{2}) = \frac{\pi}{2}$ from (i) $\rightarrow$ P.T.O.

and $f'(x) = 2 \cdot \frac{1}{\sqrt{1-x^2}}$ ✓✓

$f'(\frac{\sqrt{2}}{2}) = 2 \cdot \frac{1}{\sqrt{1-\frac{1}{2}}} = \frac{4}{\sqrt{2}} = 2\sqrt{2}$ ✓✓

Eqn: $y - y_1 = m(x - x_1)$

$y - \frac{\pi}{2} = \sqrt{2}(x - \frac{\sqrt{2}}{2})$

$y = \sqrt{2}x - 2 + \frac{\pi}{2}$ ✓✓

Q2

a) $2|x-1| = 4x-1$

$2(x-1) = 4x-1$ or $2(x-1) = 4x-1$

$2x-2 = 4x-1$ $-2x+2 = 4x-1$

$-1 = 2x$ $3 = 6x$

$x = -\frac{1}{2}$ ✓ $x = \frac{1}{2}$ ✓

CHECK NO

$x = \frac{1}{2}$ ✓

YES

③

b) $x^3 - 3x^2 - 2x + 1 = 0$

i) $a+b+c = -b/a = \frac{-2}{1} = -2$ ✓

ii) $abc = -d/a = -1$ ✓

iii) $a^2+b^2+c^2 = \frac{b^2}{a^2} = 4$ ✓

iv) $a^2+b^2+c^2 = (a+b+c)^2 - 2(ab+ac+bc)$ ✓
 $= 3^2 - 2(-2)$ ✓
 $= 13$ ✓

①

①

①

②

hard to give part marks.

c) i) In $\triangle PRT, \triangle QRT$

$\angle PTQ$ is common ✓

$\angle TPQ = \angle TRP$ ($\angle$ in altern. segment) ✓

$\angle PQT = \angle RPT$ ($\angle$ sum Δ) ✓

$\therefore \triangle PRT \sim \triangle QRT$ (equiangular) ✓

3

ii) $\frac{PT}{RT} = \frac{QT}{RT}$ (corresp sides, $\triangle PRT \sim \triangle QRT$) ✓

$\therefore PT^2 = QT \times RT$

①

Q3

a) $\int x^2(x^3-5)^5 dx$

$= \int x^2 u^5 \cdot \frac{du}{3x^2}$ ✓

$= \frac{1}{3} \int u^5 du$

$= \frac{1}{3} \cdot \frac{u^6}{6} + c$

$= \frac{1}{18} (x^3-5)^6 + c$ ✓

$u = x^3 - 5$

$\frac{du}{dx} = 3x^2$

$dx = \frac{du}{3x^2}$

1/2 for incorrect result of $\frac{1}{3}u$

relatable!

b) $\tan \alpha = \frac{m_1 - m_2}{1 + m_1 m_2}$

$\tan 45 = \frac{\frac{1}{2} - m_2}{1 + \frac{1}{2} m_2}$ ✓

$1(1 + \frac{1}{2} m_2) = \frac{1}{2} - m_2$

$\frac{3}{2} m_2 = -\frac{1}{2}$

$3 m_2 = -1$

$m_2 = -\frac{1}{3}$ ✓

Other way would be 45° the other way i.e. at 90° from $m = -\frac{1}{3}$

$\therefore m_1 = 3$ ✓

$-\frac{1}{2}$ for $m_1 + m_2$ or variants 1/6

3c

$$\frac{dT}{dt} = -k(T-c)$$

$$i) T = c + Ae^{-kt}$$

$$Ae^{-kt} = T - c \quad \textcircled{1}$$

$$\frac{dT}{dt} = 0 + -kAe^{-kt}$$

$$\frac{dT}{dt} = -k(T-c) \text{ from } \textcircled{1}$$

need to explain
that $Ae^{-kt} = T - c$
somehow
-1 if they don't

$$ii) \text{ When } t=0, T=90 \text{ Also } c=25$$

$$90 = 25 + Ae^0$$

$$65 = A$$

$$\therefore T = c + 65e^{-kt}$$

$$\text{When } t=1, T=70$$

$$70 = 25 + 65e^{-k}$$

$$45 = 65e^{-k}$$

$$\frac{45}{65} = e^{-k}$$

$$\ln \frac{45}{65} = -k$$

$$\therefore k = 0.3677$$

$$\text{so } T = 25 + 65e^{-0.3677 \dots t}$$

(iii) After

$$T = 25 + 65e^{-0.3677 \dots \times 3}$$

$$= 46.568 \text{ or } \boxed{47^\circ} \text{ (nearest } ^\circ)$$

1 for correct subs.
and calc. 0 otherwise

$\frac{1}{2}$ for $\cos^2 A = \frac{1}{2}$ (etc.)
only

$$4a. \int \cos^2 2x \, dx$$

$$\cos 2A = 2\cos^2 A - 1$$

$$\therefore \cos^2 A = \frac{1}{2}(\cos 2A + 1)$$

$$= \frac{1}{2} \int \cos 4x + 1 \, dx$$

$$= \frac{1}{2} \left(\frac{1}{4} \sin 4x + x \right) + c$$

$$= \frac{1}{8} \sin 4x + \frac{1}{2} x + c$$

$-\frac{1}{2}$ for $[-\frac{1}{2} \sin 4x]$

$\frac{1}{2}$ for leaving
off $\frac{1}{2}$ from
 $\frac{1}{2}(\cos 4x + 1)$
with next correct

$$b. i) {}^{11}C_4 = 330$$

$$ii) {}^{10}C_3 = 120$$

4c)

$$\text{Let } S(n) : (0 \times 1) + (1 \times 2) + (2 \times 3) + \dots + (n-1)n = \frac{(n-1)n(n+1)}{3}$$

SHOW TRUE FOR $S(2)$

$$\text{LHS} = (0 \times 1) + (1 \times 2)$$

$$= 2$$

$$\text{RHS} = \frac{(2-1)2(2+1)}{3}$$

$$= 2$$

$$= \text{LHS} \therefore \text{true for } S(2)$$

2/ ASSUME TRUE FOR $S(k)$

$$\text{i.e. } (0 \times 1) + (1 \times 2) + (2 \times 3) + \dots + (k-1)k = \frac{(k-1)k(k+1)}{3}$$

3/ PROVE TRUE FOR $S(k+1)$

$$\text{i.e. } (0 \times 1) + (1 \times 2) + (2 \times 3) + \dots + (k-1)k + k(k+1) = \frac{k(k+1)(k+2)}{3}$$

$$\text{LHS} = (0 \times 1) + (1 \times 2) + (2 \times 3) + \dots + (k-1)k + k(k+1)$$

$$= \frac{(k-1)k(k+1)}{3} + k(k+1) \text{ from ASSUMPTION } \textcircled{1}$$

$$= \frac{(k-1)k(k+1)}{3} + \frac{3k(k+1)}{3}$$

$$= \frac{k(k+1)[(k-1)+3]}{3}$$

$$= \frac{k(k+1)(k+2)}{3}$$

$$= \text{RHS}$$

$\therefore$ if true for $S(k)$ then true for $S(k+1)$
Since true for $S(2)$ then by the principle
of mathematical induction, $S(n)$ is true for
all positive integers $n \geq 2$.

$$4d) i) S(0, a) \quad A(x_0, -a)$$

$$\text{EQN LINE PO: } \frac{y-y_1}{x-x_1} = \frac{y_2-y_1}{x_2-x_1}$$

$$\frac{y-0}{x-0} = \frac{ap^2-0}{2ap-0}$$

$$y = \frac{ap^2}{2ap} x$$

$$y = \frac{p}{2} x$$

$$iii) m_{\text{ac}} = \frac{-a-a}{-2ap-0}$$

$$= \frac{-2a}{-2a} \cdot p$$

$$= p$$

$$m_{\text{tangent at P}} = \frac{dy}{dx}$$

$$= \frac{dy}{dp} \cdot \frac{dp}{dx}$$

$$= 2ap \cdot \frac{1}{2a}$$

$$= p$$

$$ii) A + A, y = -a$$

$$-a = \frac{p}{2} x \quad (A \text{ lies on } PO)$$

$$\therefore x = -\frac{2a}{p}$$

AS//tangent at P (equal gradients)

(1) P.O. ... then + showing also needed (1a)